

AYUSH SRIVASTAVA

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Education

University of San Francisco

M.S. in Data Science — GPA: 3.91/4.0

Coursework: Advanced Machine Learning, Deep Learning for AI, MLOps, Distributed Computing, Distributed Data Systems

Birla Institute of Technology and Science, Pilani Campus

M.Sc. Mathematics & B.E. Civil Engineering

San Francisco, CA

July 2025 - July 2026

Pilani, India

Aug 2016 - May 2021

Technical Skills

Languages: Python, SQL, C/C++

ML / DL: PyTorch, Hugging Face Transformers, Scikit-Learn, LightGBM, XGBoost, QLoRA, FAISS

Distributed & MLOps: Apache Spark, PySpark, Spark MLlib, Airflow, Docker, Kubernetes, FastAPI, MLflow

Cloud & Storage: GCP (Dataproc, Cloud Run, GCS, BigQuery), AWS Redshift, Postgres, Redis, MongoDB

Experience

Data Scientist Intern

San Francisco Bay Area, CA (Remote)

LexisNexis USA

Dec 2025 - Present

- Deployed **LLM** summarizer in **Python** via proxy API, lifting **SME pass rate from 94.74% to 100%** on a 0-3 rubric.
- Benchmarked **Claude Sonnet vs. Haiku** on latency, cost, quality across **20+** test cases, raising accuracy from **75% to 91%**.
- Extended pipeline via **Python** task generating **UIDs** for **dual-LLM calls**: summarization and span-mapping to source.
- Built batch ingestion for **100+** PDF, DOCX, HTML files using pdfplumber, docx2txt, BeautifulSoup, cutting prep by **90%**.

Senior Data Engineer

Gurgaon, India

Atria-Ingenious Insights

May 2023 - May 2025

- Modeled **20M+** records from **8+** vendors, engineering features and **30+** insights informing strategy for **6+** pharma products.
- Optimized ingestion pipelines, boosting efficiency by **40%** via automation with **SQL, Python, Matillion, AWS Redshift**.
- Led ETL and feature engineering, productizing **API wrappers** in in-house ETL tool to integrate **10+** reports into Tableau.
- Validated production pipelines with **100+** JIRA test cases, ensuring defect-free data delivery to downstream teams.

Data Engineer

Gurgaon, India

Atria-Ingenious Insights

Jul 2021 - Apr 2023

- Led team of **3** to build Tableau dashboards defining **20+** KPIs, improving sprint planning efficiency by **60%** across **4+** teams.
- Automated legacy BI pipelines and applied data reconciliation, improving reporting accuracy by **15%**.

Data Scientist Intern

Remote

Sequaretek IT Solutions Pvt. Ltd.

May 2020 - Jul 2020

- Built **VAE** and **LSTM** models in **Python** for unsupervised anomaly detection on **1M+** Active Directory event logs across **150+** predictors, flagging cyberattack patterns with **KL-divergence loss 2.23/4.60** (train/val).

Projects and Publications

ER Vital Signs Prediction | *LightGBM, GRU, PyTorch, Python*

Spring 2026

- Finished **2nd** in class competition forecasting ER patient vital signs on **14,420** observations across **1,596** patients; final submission used **30 LightGBM** models with zero tuned post-processing parameters.
- Engineered **520+** features with fold-stable selection and cross-observation aggregates; validated leakage via adversarial **AUC 0.9998** under **GroupKFold** CV by patient.
- Built two-model residual architecture with a soft-gate under **Huber loss**, combining a **GRU** sequence model and **LightGBM** ensemble; ran systematic ablations to lock final design.

Legal Holding Generation with RAG and Fine-Tuned LLMs | *Mistral 7B, QLoRA*

Jul 2025 - Aug 2025

- Fine-tuned **Mistral 7B** on LexGLUE with **4-bit QLoRA**, training **33.5M** parameters and cutting model size from **28GB** to **3.5GB**; reached cosine similarity **0.8962** on **200** test cases.
- Built **FAISS** retrieval over **Legal-BERT** 768-d embeddings, comparing **3** strategies across **1,000** LexGLUE records.

Cook Bot Cuisine Classifier | *FastAPI, Docker, MLflow, GCP, Scikit-Learn*

Fall 2025

- Served cuisine classifier over **39,774** recipes and **20** cuisines via **FastAPI** and **Docker**, reaching **78.72%** accuracy; registered in **MLflow** on **GCP** and published to **Docker Hub**.

Driver Drowsiness Detection using GRU with CNN Features | *IAPR-CVIP 2020, Springer*

Dec 2020

- Conv1D-GRU yawn detector on **AffectNet** (1M records), achieving **99.97%** val accuracy, **30 FPS** host / **23 FPS** embedded. DOI.